

Ryan Nguyen

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EDUCATION

University of Massachusetts Amherst, B.S. in Computer Science & Mathematics

Aug 2022 - May 2026

Relevant Coursework: Data Structures & Algorithms, Web Programming, Computer Systems, Algorithm Design, Artificial Intelligence, Search Engines, Combinatorics

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL

Frameworks & Tools: React, Node.js, Express.js, Django, Git, Docker, Figma, PyTorch, PostgreSQL, MongoDB, Sequelize, ESP32, Pinecone

WORK EXPERIENCES

University of Massachusetts Amherst

Amherst, MA

Undergraduate Course Assistant | CS198C Teaching Assistant & CS Tutor

Sep 2025 - Present

- Hold **4-hour** weekly office hours and explain core concepts in C programming and Unix/Linux systems.
- Monitor and respond to student questions on the course forum with an average response time under **1 hour**.
- Tutor **670+** students across CS160 (OOP), Advanced Python, and CS311 (Intro to Algorithms) through pedagogy practicum.
- Lead workshops for 250+ students to strengthen understanding of advanced C programming concepts.

GameStop Corp.

Braintree, MA

Retail Keyholder & Seasonal Worker

Jul 2024 – Jun 2025

- Processed **150+** customer transactions per week, handling **\$3,500** in cash with zero discrepancies through strict reconciliation procedures.
- Audited inventory across **15** aisles each week, identifying and resolving discrepancies within **24 hours** to maintain accurate stock levels.

PROJECTS

RemoteRide | ESP32, C++, HTML, CSS, JavaScript

Jan 2025 – Jun 2025

- Collaborated in a team of **3** to build a battery-powered toy car controlled via an ESP32 microcontroller with a custom web interface for real-time motor and servo control.
- Implemented live tracking of the car's location through the web interface, allowing users to monitor and control the vehicle remotely.
- Integrated movement control based on finger position detected by onboard camera, enhancing user interactivity.

F1 Nab | Python, CSS, React, Node.js, Pinecone, RAG, SQLite, Flask

Nov 2024 - Dec 2024

- Led a team of **4** to build a web app that embeds F1 TV into a web app, allowing users to watch races and chat with the bot simultaneously.
- Developed an interactive chatbot to provide real-time insights into F1 races, helping new fans understand terminology, strategy calls, and race events.
- Implemented a backend pipeline combining race data from Ergast API, vector embeddings with Pinecone, and Gemini API integration to deliver fast responses.

Mafia | Python, Jupyter Notebook, OpenAI API

Oct 2024 - Dec 2024

- Collaborated in a team of **3** to develop a real-time AI-driven Mafia/Werewolf game using OpenAI's language models for smooth storytelling and player interactions.
- Designed and implemented RESTful API integrations with OpenAI endpoints to enable dynamic, context-aware AI responses for real-time gameplay.
- Developed automated test cases for game logic and AI outputs to verify rule consistency and reduce unexpected model behavior.

UKart | Node.js, Express.js, SQLite, Sequelize, CSS

Sep 2024 – Dec 2024

- Led a team of **6** in building a full-stack marketplace platform for college students to buy, sell, and exchange goods securely and efficiently.
- Implemented a SQLite database with Sequelize ORM, ensuring efficient data storage and retrieval for listings and user profiles.
- Contributed to both frontend and backend, implementing key features including item listings, user profiles, and user verification.